

TASK-1

1) Chatgpt(OpenAI)

ChatGPT is a generative artificial intelligence chatbot developed by OpenAI.

Originally released in November 2022, the product uses large language models -

specifically generative pre-trained transformers (GPTs)—to generate text,

speech, and images in response to user prompts.

* history , evolution of models ,ideal use cases & capabilities->

-GPT-3.5- first model used in gpt devloped in Nov 2022

-GPT-4 -larger then gpt3.5 developed in Mar 2023

-GPT-4o - Capable of processing text, image, audio, and video, faster and more capable than GPT-4.

Was accused of being particularly sycophantic.The variant GPT-4o mini replaced GPT-3.5 in the July 2024

-o1 -An early reasoning model (one that can reason through a problem before answering). Slower, but can solve more complex problems, particularly in math, coding and science .developed in 2024

-GPT-4.5 -large gpt model developed in 2025

-GPT-4.1 - Primarily intended for coding. Particularly long context window of 1 million tokens (around 3,000 pages of text).-2025

-GPT-o3 - Can analyze images and has access to tools such as web search and a Python code execution environment -2025

-GPT-5 - can either answer quickly like earlier GPT models or reason before answering like o1 or o3.Rather than a single model, it consists of a network of models with different levels of capability, with a router selecting one based on the complexity of the task-Aug 2025

-GPT-5.1 -Allows users to select alternative personalities, such as friendly, efficient-2025

-GPT-5.2 - Released in a context of competitive pressure from Google with Gemini.-2025

-GPT-5.3 -OpenAI released only two GPT-5.3 models: GPT-5.3-Codex (for coding) in February

2026 and GPT-5.3 Instant (for quick answers) in March 2026

-GPT-5.4- Focused on enterprise use, with an improved ability to autonomously operate

computers and interact with external tools. Needs fewer tokens to solve problems

-GPT-5.5- Designed for complex, multi-step professional work including agentic coding

and computer use.OpenAI released a variant for cyber defense named "GPT-5.5-Cyber"

*Limitations->

-> A 2023 analysis estimated that ChatGPT hallucinates around 3% of the time.

->A 2025 analysis by The Washington Post estimated that ChatGPT started its responses with words like "yes" or "correct" almost 10 times more often than "no" or "wrong"

->May exhibit bias present in training data, if the data is not cleaned well or has some

wrong info then the model will give wrong outputs.

->Reasoning ability varies across different model versions.

using right model for particular work like for coding gpt5.3 codex is used

*Strengths ->

->Can explain long files pdfs in short in no time.

->supports multimodal input like texts,media ,audio,video

->uses the memory feature to summary the whole memory of the the chat.

->Can automate repetitive tasks such as report writing, email

drafting, and documentation.

->Supports tool usage such as web search, code execution, and file analysis.

->using right model for right use maximizes the output and minimizes tokens and time.

*Which Model is Best Suited for Specific Tasks->

- General Chatting--> GPT-4o / GPT-5
- Coding--> GPT-4.1, GPT-5.3 Codex, GPT-5.5
- Mathematics--> o1, o3, GPT-5
- Research & Analysis--> GPT-5, GPT-5.5
- Image Understanding--> GPT-4o, o3
- Long Documents--> GPT-4.1
- Business Automation--> GPT-5.4
- Creative Writing--> GPT-4.5, GPT-5
- Educational Assistance--> GPT-4o, GPT-5

*Effective Prompting Techniques and Best Practices->

1.be specific what you want

ex:

instead of :

explain java

use:

Explain Java OOP concepts with real-world examples suitable for a second-year engineering student.

2.provide context

ex:

instead of:

write email

use:

Write a professional email to a recruiter requesting an

internship extension by two weeks.

3.define output format

like if u want the output in pdf having points: definition ,tech , steps, conclusion

4.use step by step instructions

5.role prompting- provide role abt the info u want

6. asks for example

7. do not fully rely on gpt ,cross check your answers

*How to Maximize Productivity and Output Quality->

dos-->Provide clear and detailed prompts.

Break complex tasks into smaller subtasks.

Use follow-up questions to refine responses.

Ask for explanations, examples, and alternatives.

Verify critical information independently.

Use uploaded files/documents for better context.

donts-->Trust outputs blindly without verification.

Expect perfect accuracy in factual or recent information.

Share sensitive personal or confidential information.

references- <https://en.wikipedia.org/wiki/ChatGPT>

2)Claude(by anthropic)

Claude is a family of large language models developed by Anthropic, an AI research company founded in 2021 by former OpenAI researchers including Dario Amodei and Daniela Amodei. Claude was designed with a strong focus on AI safety, helpfulness, honesty, and harmlessness. Anthropic uses a training approach called Constitutional AI, where the model learns to follow a set of principles

instead of relying solely on human feedback.

Claude is widely used for:

- Research and analysis
- Long-document processing
- Programming assistance
- Writing and editing
- Business workflows

*History, Evolution, Model Capabilities & ideal use cases->

1)Claude 1(2023)-

Anthropic's first major public model.

Weaker reasoning than GPT-4.

Less capable in coding.

2)Claude 2 (2023)- & 2.1(reduced hallucinations)

Larger context window.

Better reasoning, coding support.

Improved document understanding.

-Could process approximately 100,000 tokens,
allowing users to analyze very large documents
and research papers.

3)Claude 3 Haiku-

Very fast responses.

Low cost.

Suitable for simple tasks.chatbots,basic content generation,quick
summaries.

4)Claude 3 Sonnet-Good reasoning.

Good coding performance.

Fast response times. best for Software development,Research

5) Claude 3 Opus

Most powerful model in Claude 3 series.

Advanced reasoning.

Strong coding.

High-quality writing.

Better problem-solving. Best For Complex analysis , Research projects

, Advanced coding

6) Claude 3.5 Sonnet (2024)

One of Anthropic's most successful releases.

Major coding improvements.

Better reasoning.

Faster responses.

Improved instruction following. it is the best among coding assistants available.

7) Claude 4 Sonnet

Improved coding.

Better reasoning.

More accurate tool usage.

Better software engineering performance.

8) Claude 4 Opus

Most advanced Claude model.

Strongest reasoning capability.

Better planning.

Long multi-step task execution.

Advanced agent-like workflows.

*Strengths and Ideal Use Cases->

1) Claude is known for handling extremely long documents-like research papers, books, legal docs etc

2)High-Quality Writing-

proff reports,essays,documentation.

Many users consider Claude's writing style more natural and human-like than most competitors.

3)Strong Coding Assistance-generate code,debug prblms,explain algo,review architecture.

particularly strong for java, python, js,typescript

4)Research & Analysis-excellent at summarizzation,comparative analysis,identifying trends.

5)works on constitutionalAI to reduce bias, harmful outputs,unsafe responses.

6) Why choose claude over chatgpt?-->Claude often provides better solutions,
with fewer errors and more attention to detail.it seeks to understand the intention
behind your request and offers the most optimized solution.

*Limitations-->

->Hallucinations (Generating Incorrect Information)-

Anthropic's own help center acknowledges that Claude may produce incorrect or misleading responses, including fabricated citations and outdated information. The company explicitly warns users not to rely on Claude as the sole source of truth for high-stakes decisions.

->Less Reliable for Real-Time Information-

Anthropic states that Claude may become confused when asked about current events because

it may not have been trained on the latest information.

-> Sometimes Overly Cautious or Refuses Benign Requests-

Anthropic's Constitution explicitly discusses the risk of Claude being "overly cautious" and refusing reasonable requests. The company acknowledges that excessive refusals reduce usefulness.

-> Mathematical and Logical Errors Can Still Occur-

Recent research shows that language models can sometimes perform correct reasoning steps but still produce incorrect final answers. Claude models were observed making such mistakes on specialized reasoning benchmarks.

-> Safety Restrictions May Limit Flexibility-

Claude prioritizes safety and responsible AI use.

*Differences Between Claude Versions-

Claude 1-->	Basic conversational AI
Claude 2-->	Long-document understanding
Claude 2.1-->	Better accuracy, fewer hallucinations
Claude 3 Haiku-->	Fastest and cheapest
Claude 3 Sonnet-->	Balanced performance
Claude 3 Opus-->	Strong reasoning and writing
Claude 3.5 Sonnet-->	Excellent coding
Claude 3.7 Sonnet-->	Hybrid reasoning
Claude 4 Sonnet-->	Advanced software engineering
Claude 4 Opus-->	Best reasoning and enterprise workflows

*Which Claude Model is Best for Specific Task->

General Chat--> Claude 3 Sonnet / 4 Sonnet

Coding--> Claude 3.5 Sonnet, 4 Sonnet

Large Documents--> Claude 4 Opus

Research--> Claude 4 Opus
Writing--> Claude 3 Opus, 4 Opus
Summarization--> Claude 4 Opus
Business Analysis--> Claude 4 Opus
Customer Support--> Claude 3 Haiku

***Effective Prompting Techniques-->**

- >clear context
- >specify output
- >use less tokens as much as possible
- >use role prompting
- >step b y step requests

3)Gemini(Google)-->

Gemini is a family of multimodal AI models developed by Google DeepMind.

It was introduced in December 2023 as Google's successor to the PaLM model family and is designed to understand and generate text, images, audio,

video, and code.

Gemini powers many Google products, including:

Google Search AI features

Gemini App

Google Workspace

Android AI features

Google Cloud AI services

A major goal of Gemini is to provide a single model architecture capable of handling multiple types of information (multimodal AI).

***History, Evolution , Model Capabilities & use cases-->**

1)Gemini 1.0(dec 2023)

Google launched three versions:

i>Gemini Nano

Smallest model designed for on-device tasks.

-Features

Runs directly on smartphones.

Optimized for low latency.

Supports Android AI features.

-Best For

Mobile AI assistants

Smart replies

On-device summarization

ii>Gemini Pro

Mid-sized model.

-Features

General-purpose AI model.

Replaced Google's earlier PaLM-based chatbot.

Strong reasoning and coding abilities.

-Best For

Chat applications

Content creation

Productivity tasks

iii>Gemini Ultra

Largest and most capable Gemini 1.0 model.

-Features

Advanced reasoning.

Strong coding performance.

Better benchmark performance than previous Google models.

-Best For

Research

Complex problem solving

Professional applications

2) Gemini 1.5 Pro (2024)

One of Google's biggest breakthroughs.

-Major Improvement

Introduced an extremely large context window.

-Capabilities

Up to 1 million tokens initially.

Later expanded to 2 million tokens.

-Users could analyze:

Hundreds of pages of PDFs

Large codebases

Long research reports

Entire books

-Best For

Research

Legal documents

Enterprise knowledge bases

Software engineering

3) Gemini 1.5 Flash (2024)

Fast and lightweight version.

-Features

Lower cost.

Faster responses.

Optimized for high-volume applications.

-Best For

Chatbots

Customer support

Quick summaries

Real-time interactions

4) Gemini 2.0 (2025)

Introduced more agentic capabilities.

-Improvements

Better reasoning.

Improved tool usage.

Better integration with Google ecosystem.

Enhanced multimodal understanding.

-Best For

Research

Workflow automation

Productivity applications

5) Gemini 2.5 Pro (2025)

Google's flagship reasoning model.

-Features

Advanced reasoning.

Strong coding performance.

Improved mathematics.

Better long-context understanding.

Agentic workflows.

-Best For

Software development

Research analysis

Strategic planning

Enterprise applications

*Strengths->

->1. Multimodal Understanding

Gemini was designed from the beginning as a multimodal model.

Can Process--Text ,Images ,Audio ,Video ,Code.

->Extremely Large Context Window

Gemini 1.5 Pro became famous for its long-context capability.

Can analyze:Entire books ,Large repositories ,Research collections without extensive chunking.

->Strong Google Ecosystem Integration

Gemini integrates with:Gmail ,Docs ,Sheets, Drive ,Search ,Android

->Coding Assistance-

code generation,debugging,documnetation,refactoring

*Limitations->

->hallucinations-

Google states that Gemini may sometimes provide inaccurate responses and encourages users to verify important information.

->Early Image Generation Problems-

Google paused Gemini's image generation feature after users discovered historically inaccurate image outputs

->Mathematical Error

->Over-Reliance on Prompt Quality

Gemini often performs much better when given:Detailed instructions ,
Clear context ,Structured tasks

*Differences Between Gemini Versions->

Model	Main Strength
Gemini Nano->	On-device AI
Gemini Pro->	General-purpose assistant
Gemini Ultra->	Advanced reasoning
Gemini 1.5 Pro->	Massive context window
Gemini 1.5 Flash->	Fast and low cost
Gemini 2.0->	Better agentic capabilities
Gemini 2.5 Pro->	Strong reasoning and coding

*Which Gemini Model is Best for Specific Tasks->

Task	Recommended Model
General Chat->	Gemini Pro / Gemini 2.5 Pro
Coding->	Gemini 2.5 Pro
Research->	Gemini 2.5 Pro
Long Documents->	Gemini 1.5 Pro
Business Productivity->	Gemini 2.5 Pro

Mobile AI Gemini Nano-> Gemini Nano
Fast Responses-> Gemini 1.5 Flash
Image Generation-> Gemini 2.0 Flash Image Generation / Gemini App
(Imagen-powered)
Image Understanding-> Gemini 2.5 Pro
Multimodal Tasks (Text + Image + Audio + Video)-> Gemini 2.5 Pro

*Image Generation Capabilities->

Google integrates image generation through its Imagen family of models, which work alongside Gemini.

-Features

Text-to-image generation

Image editing

Background replacement

Object insertion and removal

Style transformation

High-quality photorealistic image

4)Kimi(Moonshot AI)-->

Kimi is a large language model and AI assistant developed by Moonshot AI, a Chinese artificial intelligence company founded in 2023.

Kimi gained popularity because of its ability to process extremely long documents and large context windows, making it particularly useful for research, document analysis, and knowledge-intensive tasks.

Unlike many early AI assistants that focused primarily on conversation, Kimi was designed to help users work with:

Research papers

Books

Technical documentation

Large PDFs

Long conversations

*History, Evolution & Model Capabilities->

1)Early Kimi Assistant (2023)

Better handling of large documents than many competitors at the time.

The first public Kimi assistant focused on:

-Features

Conversational AI

Document understanding

Long-context processing

Research assistance

-Best Use Cases

Academic research

Report analysis

Knowledge management

2)Kimi K1 Series (2025)

Introduced stronger reasoning capabilities.

-Improvements

Better logical reasoning.

Improved mathematics.

Enhanced coding performance.

Better multilingual support.

-Best For

Programming

Problem solving

Research analysis

Education

3)Kimi K2 (2025)

One of Moonshot AI's most advanced models.

-Features

Advanced reasoning.

Strong coding capabilities.
Long-context understanding.
Improved instruction following.
Better tool usage.

-Capabilities

Software engineering

Research projects

Data analysis

Knowledge extraction

-Best Use Cases

Enterprise research

Coding assistance

Large document analysis

Technical writing

*Strengths and Ideal Use Cases->

->Exceptional Long Context Processing-

Kimi's biggest strength is handling large amounts of information.

->Strong Research Assistance-

Kimi performs well at: Summarization , Information extraction

->Coding Support

->4. Multilingual Capabilities

Kimi performs particularly well in: Chinese , English

->5. Cost Efficiency-

Compared with some premium AI services,

Kimi has often been positioned as a more cost-effective

*Limitations-->

->Hallucinations

->Smaller Global Ecosystem-Fewer third-party integrations,

Smaller developer ecosystem , Less global adoption

->Image Generation Limitations-

Kimi is primarily a language and reasoning model.
For creative image generation, users often prefer other AI.

*Differences Between Kimi Versions-->

Model	Main Strength
Early Kimi-->	Conversational AI and document understanding
Kimi Long Context-->	Large-document analysis
Kimi K1-->	Improved reasoning and coding
Kimi K2-->	Advanced reasoning, coding, and research

*Which Kimi Model is Best for Specific Tasks?

for most of the tasks it is found that kimi k2 works very well

5)DeepSeek (DeepSeek AI)-->

DeepSeek is a family of large language models developed by DeepSeek AI, a Chinese artificial intelligence company founded in 2023.

DeepSeek gained worldwide attention because it delivered performance comparable to leading AI models while being significantly more cost-efficient.

-The company focuses heavily on:

Reasoning

Coding

Mathematics

Research

Open-weight AI models

Unlike many competitors that keep their models fully closed, DeepSeek has released several open-weight models, allowing researchers and developers to study and build upon them.

*History, Evolution & Model Capabilities-->

1)DeepSeek Coder (2023)

One of DeepSeek's earliest successful model families.

-Features

Specialized for programming tasks.

Trained on large amounts of source code.

Supported multiple programming languages.

-Capabilities

Code generation

Debugging

Code explanation

Documentation

Best Use Cases

Software development

Learning programming

Technical interviews

DeepSeek LLM (2023)

2)General-purpose language model.

-Features

Conversational AI

Research assistance

Content generation

Multilingual support

-Best Use Cases

Chat applications

Writing assistance

Educational support

3)DeepSeek V2 (2024)

Major advancement in efficiency.

-Improvements

Better reasoning.

Faster inference.
Lower operating costs.
Improved multilingual support.
Notable Achievement

DeepSeek V2 attracted attention because it delivered strong benchmark performance while requiring fewer computational resources than many competitors.

4)DeepSeek V3 (2024)

Flagship foundation model.
-Improvements
Better reasoning.
Improved coding.
Better instruction following.
Strong multilingual capabilities.
-Features
Large-scale training.
High-quality responses.
Better context understanding.

The model that made DeepSeek globally famous.
R1 was designed specifically for reasoning tasks.

-Features
Step-by-step problem solving.
Mathematical reasoning.
Scientific reasoning.
Advanced coding support.
-Why It Became Popular
Many evaluations showed DeepSeek R1 competing with or approaching the performance of leading reasoning models while remaining significantly cheaper.

*Strengths-->

-->Strong Reasoning Ability-

DeepSeek R1 is particularly known for reasoning.

-->Excellent Coding Performance

-->Cost Efficiency-

One of DeepSeek's biggest advantages.

-->Open-Weight Models-

Unlike many competitors, DeepSeek has released open-weight models.

*Limitations-->

-->Hallucination

-->Limited Ecosystem Compared to OpenAI and Google

-->Inconsistent Real-World Knowledge

-->Limited Enterprise Adoption-

Although growing rapidly, DeepSeek is still less widely adopted in enterprises compared to major Western AI platforms.

-->Image Generation Not a Core Strength

*Differences Between DeepSeek Versions-->

Model	Main Strength
DeepSeek Coder-	Programming and software development
DeepSeek LLM-	General-purpose AI
DeepSeek V2-	Efficiency and cost optimization
DeepSeek V3-	Better reasoning and coding
DeepSeek R1-	Advanced reasoning and mathematics

*Which DeepSeek Model is Best for Specific Tasks?-->

Task	Recommended Model
General Chat-	DeepSeek V3

Coding- DeepSeek Coder / DeepSeek V3
Mathematics- DeepSeek R1
Logical Reasoning- DeepSeek R1
Research- DeepSeek V3
Competitive Programming- DeepSeek R1
Educational Use- DeepSeek R1
Document Analysis- DeepSeek V3
Image Analysis- DeepSeek V3 (where supported)

6)Grok(xAI)-->

Grok is a family of artificial intelligence models developed by xAI, an AI company founded by Elon Musk in 2023.

The name "Grok" comes from Robert Heinlein's science-fiction novel *Stranger in a Strange Land*, where "grok" means to understand something deeply and intuitively.

Grok was created to compete with ChatGPT, Claude, Gemini, and DeepSeek. One of its unique features is its integration with X (formerly Twitter), allowing it to access and analyze real-time social media discussions.

Grok is designed to:

- Answer questions
- Assist with coding
- Perform reasoning tasks
- Analyze real-time information
- Generate and edit images
- Support research and productivity workflows

***History, Evolution & Model Capabilities-->**

1)Grok-1 (2023)

The first public Grok model.

-Features

Conversational AI.

Integrated with X platform.

Access to real-time social media content.

-Best Use Cases

News monitoring.

Trend analysis.

General conversation.

2)Grok-1.5 (2024)

Major upgrade over Grok-1.

-Improvements

Better reasoning.

Improved coding performance.

Larger context window.

Enhanced mathematical abilities.

-Best Use Cases

Programming.

Research.

Technical analysis.

3)Grok-2 (2024)

Significant performance improvements.

-Features

Better instruction following.

Improved reasoning.

Better image understanding.

Enhanced coding abilities.

-Best Use Cases

Software development.

Research assistance.

Educational support.

4)Grok-3 (2025)

xAI's flagship model.

-Major Improvements

Advanced reasoning.

Strong mathematics performance.

Better coding capabilities.

Improved multimodal understanding.

Faster response generation.

-Best Use Cases

Coding

Research

Data analysis

Content creation

Enterprise workflows

* Strengths and Ideal Use Cases-->

-->1. Real-Time Information Access-

If a major technology product launches today,

Grok can often discuss current public reactions much faster than models relying only on training data.

-->Strong Coding Assistance

-->Advanced Reasoning

-->Multimodal support

*Limitations-->

-->social media bias-

Because Grok heavily leverages information from X, it can sometimes be influenced

-->Less Mature Enterprise Ecosystem compared to other AI's

-->Variable Accuracy on Niche Topics

Grok may sometimes struggle with:

Highly specialized academic topics
Rare technical domains
Deep scientific research

***Differences Between Grok Versions-->**

Model	Main Strength
Grok-1	Real-time information and social media integration
Grok-1.5	Improved reasoning and coding
Grok-2	Better multimodal and coding capabilities
Grok-3	Advanced reasoning, coding, and multimodal AI

***Grok 3 is the based model among all , it can do everything with ease**

***Ask for current information**

7) Antigravity (Google)-->

Antigravity is an AI-powered, agent-first software development platform developed by Google. Unlike ChatGPT or Gemini, which mainly answer questions,

Antigravity allows AI agents to write code, run code, test applications, debug errors,

use browsers, and interact with terminals autonomously. It is primarily designed for

software engineers and development teams.

-Models Used in Antigravity

Antigravity is not a model itself. It provides access to multiple AI models.

-Main Reasoning Models

Gemini 3.1 Pro (High)

Gemini 3.1 Pro (Low)

Gemini 3 Flash

Gemini 3.5 Flash

Claude Sonnet 4.6 (Thinking)

Claude Opus 4.6 (Thinking)

GPT-OSS-120B

-Additional Internal Models

Nano Banana Pro 2 (Image generation/UI mockups)

Gemini 2.5 Pro UI Checkpoint (Browser control)

Gemini 2.5 Flash (Context summarization)

Gemini 2.5 Flash Lite (Codebase search)

*Strengths-->

1. Autonomous Coding

Can:

Generate code

Debug applications

Refactor projects

Create documentation

Run tests automatically

2. Multi-Agent Workflow

Different agents can work simultaneously on:

Frontend

Backend

Testing

Documentation

3. Browser & Terminal Access

Agents can:

Open websites

Execute commands

Verify outputs

Test applications automatically

4. Artifacts System

Provides:

Task plans

Screenshots

Reports

Browser recordings

to show exactly what the AI agent did.

5.can choose model as per work type

*Ideal Use Cases-->

Full-stack development

Enterprise software projects

Automated testing

DevOps workflows

Large coding projects

*Limitations-->

1. Requires Human Supervision

AI agents can still make mistakes and may perform unintended actions.

2. High Resource Consumption

Complex projects consume significant reasoning quota and compute resources.

3. Learning Curve

Developers must understand:

Agent orchestration

Workflow management

Multi-agent systems

4. Model Restrictions

Some enterprise accounts have access only to Gemini models, while certain third-party models may not be available.

5. Not Suitable for General Users

Primarily focused on software engineering rather than general-purpose AI

assistance.

6. has many less tokens to operate on.

8)Google Flow-->

Google Flow is an AI-powered filmmaking and video creation tool developed by Google.

It helps users create cinematic videos using natural language prompts and Google's generative AI models. Flow combines video generation, scene creation, camera controls, and storytelling into a single platform.

It was introduced at Google I/O 2025 and is aimed at creators, filmmakers, marketers, and content producers.

***Models Used in Google Flow-->**

Flow is powered by Google's generative AI models:

1. Veo

Google's video generation model.

Converts text prompts into high-quality videos.

Generates realistic motion and cinematic scenes.

2. Imagen

Google's image generation model.

Creates characters, environments, and visual assets.

3. Gemini

Helps with prompt understanding, scene planning, and storytelling.

Assists in generating narratives and creative ideas.

***Key Features-->**

1. AI Video Generation

Create videos directly from text descriptions.

2. Scene Creation & Editing

Users can:

Add new scenes

Modify existing scenes

Change backgrounds

Adjust camera angles

3. Character Consistency

Flow can maintain the same character across multiple scenes, which is useful for storytelling.

4. Cinematic Controls

Provides controls such as:

Camera movement

Zoom

Pan

Scene transitions

5. Storytelling Support

Helps creators build complete narratives instead of generating isolated clips.

* Strengths-->

-Content Creation

-YouTube videos

-Marketing content

-Advertisements

-Educational videos

-Filmmaking

-Storyboarding

-Concept visualization

-Short-film creation

-Creative Design

- Animation concepts
- Visual storytelling
- Social media content

*Limitations-->

1. Video Length Restrictions

Long movies still require editing and multiple generated clips.

2. AI Artifacts

Generated videos may occasionally contain:

Unrealistic movements

Visual inconsistencies

Object distortions

3. Prompt Dependency

Output quality depends heavily on prompt quality.

4. Compute Intensive

Video generation requires significant computing resources.

5. Limited Human Creativity

Although powerful, it cannot fully replace professional filmmakers and editors.

<--SUMMARY OF THE DOC-->

As part of this task, I explored and studied major AI platforms including ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google), Kimi (Moonshot AI), DeepSeek, Grok (xAI), Antigravity, and Google Flow. The objective was to understand their evolution, capabilities, strengths, limitations, ideal use cases, and practical applications.

----Key Learnings----

---> ChatGPT (OpenAI)

- * Most balanced AI platform for coding, reasoning, content creation, and productivity.
- * Strong multimodal capabilities through text, image, audio, and document processing.
- * GPT-5 combines fast responses with advanced reasoning.
- * Suitable for students, developers, researchers, and business users.

---Claude (Anthropic)

- * Excellent for long-document analysis and report generation.
- * Produces highly structured and natural-sounding responses.
- * Strong coding and research capabilities.
- * Known for its Constitutional AI safety approach.

---Gemini (Google)

- * Strong multimodal platform with support for text, images, audio, and video.
- * Deep integration with Google Workspace and ecosystem.
- * Large context windows make it effective for document analysis and research.
- * Supports image and video generation through Imagen and Veo.

--- Kimi (Moonshot AI)

- * Specialized in long-context processing and research assistance.
- * Effective for academic work, literature reviews, and document analysis.
- * Strong multilingual capabilities, particularly in Chinese and English.

---DeepSeek

- * Strong reasoning, mathematics, and coding performance.
- * DeepSeek R1 excels in step-by-step problem solving.
- * Cost-efficient and attractive for developers and researchers.
- * Open-weight models support customization and research.

---Grok (xAI)

- * Unique access to real-time information and social media trends.
- * Strong reasoning and coding capabilities.
- * Useful for market research, news analysis, and trend monitoring.

---Antigravity (Google)

- * Agent-based software development platform.
- * Automates coding, testing, debugging, and documentation.
- * Supports multi-agent workflows for large-scale software projects.

---Google Flow

- * AI-powered video creation and filmmaking platform.
- * Uses Gemini, Veo, and Imagen models.
- * Enables creation of cinematic videos from text prompts.

-----Comparative Analysis-----

Platform	Main Strength	Best Use Case
ChatGPT	Balanced performance	Coding, learning,
productivity		
Claude	Long-document analysis	Research, report writing
Gemini	Multimodal capabilities	Productivity, media,
research		
Kimi	Long-context understanding	Academic research

DeepSeek	Reasoning and coding	Mathematics,
programming		
Grok	Real-time information	Trend analysis, news
Antigravity	AI agents for development	Software engineering
Google Flow	AI video generation	Content creation

-----Recommendations-----

-- For Coding

1. Claude
2. DeepSeek
3. ChatGPT

--For Research and Documentation

1. Claude
2. Gemini
3. Kimi

-- For Learning and Education

1. ChatGPT
2. Gemini
3. DeepSeek

-- For Real-Time Information

1. Grok
2. Gemini
3. ChatGPT (with web access)

-- For Content Creation

1. ChatGPT
2. Gemini
3. Claude

-- For Image and Video Generation

1. Google Flow
2. Gemini
3. ChatGPT

-- For Enterprise Software Development

1. Antigravity
2. ChatGPT
3. Claude

----- Practical Observations-----

During practical evaluation, no single AI model proved best in every category.

- * ChatGPT provided the most balanced overall experience.
- * Claude generated the highest-quality reports and document summaries.
- * Gemini demonstrated excellent multimodal and ecosystem integration.
- * Kimi performed exceptionally well on large documents and research tasks.
- * DeepSeek delivered strong reasoning and coding performance at lower cost.
- * Grok was valuable for accessing current information and social trends.
- * Antigravity showcased the future of AI-assisted software development through autonomous agents.
- * Google Flow highlighted the growing role of AI in creative video

production.

Prompt quality significantly influenced output quality across all platforms. Detailed prompts with clear objectives consistently produced better results than vague instructions.

----- Conclusion-----

The AI landscape is rapidly evolving, with each platform specializing in different areas.

ChatGPT remains the most versatile general-purpose assistant, Claude excels in research

, coding & writing, Gemini leads in multimodal capabilities, DeepSeek shines in reasoning and coding,

Kimi specializes in long-context research, Grok offers real-time awareness, Antigravity focuses on agent-based software development, and Google Flow demonstrates advanced AI-powered video generation.

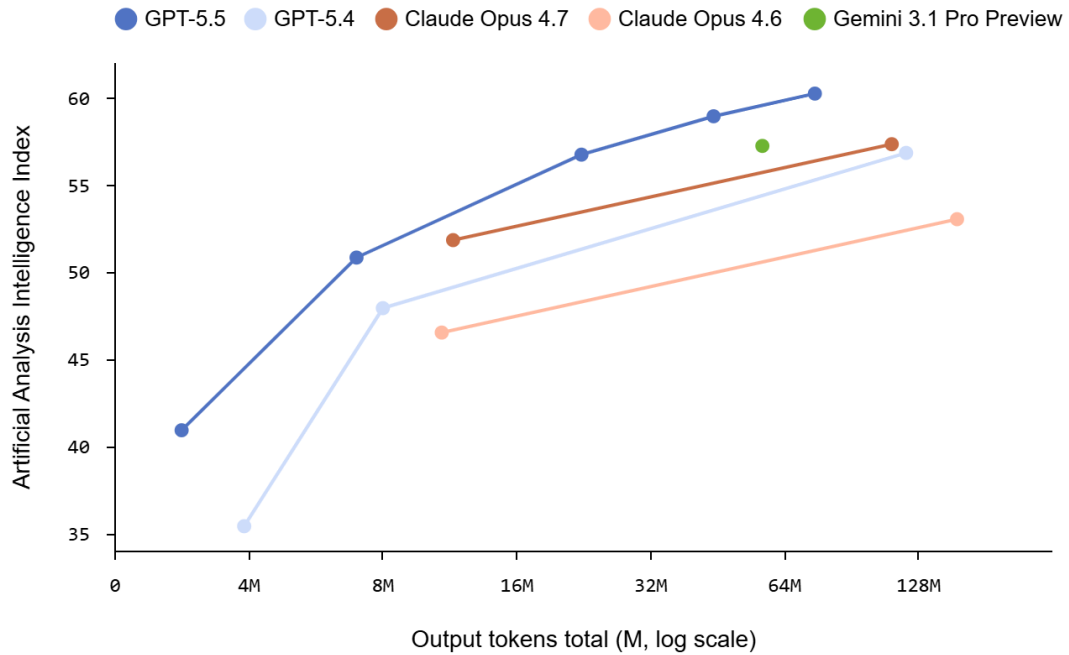
Selecting the right AI model depends on the specific task, workflow requirements, and desired outcomes.

Understanding the strengths and limitations of each platform enables more effective and productive use of AI technologies.

References - wikipedia,google,claude online documentation ,chatgpt website, reddit

	Claude 3 Opus	Claude 3 Sonnet	Claude 3 Haiku	GPT-4	GPT-3.5	Gemini 1.0 Ultra	Gemini 1.0 Pro
Undergraduate level knowledge <i>MMLU</i>	86.8% 5-shot	79.0% 5-shot	75.2% 5-shot	86.4% 5-shot	70.0% 5-shot	83.7% 5-shot	71.8% 5-shot
Graduate level reasoning <i>GPQA, Diamond</i>	50.4% 0-shot CoT	40.4% 0-shot CoT	33.3% 0-shot CoT	35.7% 0-shot CoT	28.1% 0-shot CoT	—	—
Grade school math <i>GSM8K</i>	95.0% 0-shot CoT	92.3% 0-shot CoT	88.9% 0-shot CoT	92.0% 5-shot CoT	57.1% 5-shot	94.4% Maj1@32	86.5% Maj1@32
Math problem-solving <i>MATH</i>	60.1% 0-shot CoT	43.1% 0-shot CoT	38.9% 0-shot CoT	52.9% 4-shot	34.1% 4-shot	53.2% 4-shot	32.6% 4-shot
Multilingual math <i>MGSM</i>	90.7% 0-shot	83.5% 0-shot	75.1% 0-shot	74.5% 8-shot	—	79.0% 8-shot	63.5% 8-shot
Code <i>HumanEval</i>	84.9% 0-shot	73.0% 0-shot	75.9% 0-shot	67.0% 0-shot	48.1% 0-shot	74.4% 0-shot	67.7% 0-shot
Reasoning over text <i>DROP, F1 score</i>	83.1 3-shot	78.9 3-shot	78.4 3-shot	80.9 3-shot	64.1 3-shot	82.4 Variable shots	74.1 Variable shots
Mixed evaluations <i>BIG-Bench-Hard</i>	86.8% 3-shot CoT	82.9% 3-shot CoT	73.7% 3-shot CoT	83.1% 3-shot CoT	66.6% 3-shot CoT	83.6% 3-shot CoT	75.0% 3-shot CoT
Knowledge Q&A <i>ARC-Challenge</i>	96.4% 25-shot	93.2% 25-shot	89.2% 25-shot	96.3% 25-shot	85.2% 25-shot	—	—
Common Knowledge <i>HellaSwag</i>	95.4% 10-shot	89.0% 10-shot	85.9% 10-shot	95.3% 10-shot	85.5% 10-shot	87.8% 10-shot	84.7% 10-shot

Artificial Analysis Intelligence Index



GDPval Knowledge work tasks

